



Advanced Control Solutions Towards Resource Efficient Production

ACTREP — Final Technical Summary

Project number	09I03-03-V04-00636
Implementation period	September 2024 – August 2026
Principal investigator	Ing. Michaela Horváthová, PhD
Host institution	UIAM, Faculty of Chemical and Food Technology, Slovak University of Technology in Bratislava
Report status	Public technical summary; formal agency reporting scheduled for September 2026

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Prepared in August 2026

Executive Summary

ACTREP developed, certified, implemented and disseminated advanced model predictive control (MPC) solutions aimed at resource-efficient industrial production. The project combined learning-based modelling and control, robust predictive control with formal guarantees, solver-free approximated MPC, and practical implementation on laboratory and resource-constrained platforms.

The project produced peer-reviewed journal and conference publications, experimentally validated control solutions, publicly documented MATLAB implementations, and contributions to the established MPTplus toolbox. It also strengthened international cooperation with leading European research partners and supported a follow-up APVV grant application.

Overall result

The scientific, publication, experimental and open-source objectives of WP1–WP4 were achieved. The official administrative final report will be submitted according to the grant agency’s reporting schedule in September 2026.

1. Project Objectives

- Develop novel energy-efficient control methods for industrial systems.
- Provide theoretical certification of stability, feasibility and robustness.
- Increase implementability on embedded and computationally constrained hardware.
- Validate the proposed approaches numerically and experimentally.
- Disseminate publications, software, tutorials and project information through open-science channels.

2. Work Package Results

WP1 — Development of Novel and Improved Control Methods

WP1 established the methodological basis of the project through a comprehensive literature review, algorithm development, simulation studies and international research cooperation.

- Completed a comprehensive review of energy-efficient, robust, learning-based and approximated MPC methods.
- Developed stochastic approximated MPC concepts, including recursive random-shooting formulations and variance-adaptive sampling.
- Validated the early algorithms through simulations and laboratory experiments.
- Established research cooperation supporting subsequent joint conference publications.

WP2 — Certification of the Control Methods

WP2 addressed safe use of data-driven control through feasibility, stability and robustness certification, complemented by public dissemination.

- Developed a supervised-learning framework for robust predictive control with certified feasibility and stability.
- Demonstrated substantial memory reduction compared with explicit and tube-based MPC benchmarks.
- Published the results in Neurocomputing, volume 671 (2026), article 132637.
- Delivered a presentation for the general public explaining the project objectives and results.

WP3 — Increased Implementability

WP3 reduced the computational burden of advanced MPC and translated the algorithms into reproducible implementations suitable for constrained hardware.

- Developed a compact solver-free variance-adaptive approximated MPC implementation.
- Used adaptive sampling covariance to balance exploration after significant control changes and focused sampling near steady operation.
- Included a stabilizing support controller as a feasible fallback.
- Released MATLAB code, functional tests, documentation, a quadrotor example and citation metadata.
- Presented the internationally coauthored variance-adaptive method at ECC 2026.

WP4 — Experimental Implementations and Tool Development

WP4 demonstrated practical applicability through experimental studies and delivered publicly accessible software and documentation.

- Validated deep Koopman MPC experimentally on a laboratory pasteurization unit.
- Demonstrated improved tracking performance and reduced energy use relative to a conventional benchmark.
- Published and presented a deep Koopman economic MPC study for resource-efficient pasteurization.
- Implemented time-varying cross-section tube MPC in the open-source MPTplus toolbox, including automated synthesis, examples and a public tutorial.
- Released the ACTREP variance-adaptive MPC package under the MIT licence.

3. Deliverables and Milestones

ID	Deliverable	Status
D1.1	Literature review report	Completed
D1.2	Conference publication on algorithm design and simulations	Completed
D1.3	International research cooperation	Completed
D2.1	Journal publication on certification	Published
D2.2	Presentation for the general public	Completed
D2.3	Interim implementation report	Completed; retained in project records
D3.1	Optimized implementations for constrained hardware	Completed
D3.2	International conference publication and presentation	Published and presented
D4.1	Journal publication with practical implementation	Published
D4.2	Conference publication and presentation	Published and presented
D4.3	Open-source tool package with documentation and tutorials	Completed
D4.4	Grant application and institutional notification	Completed through institutional APVV submission
D4.5	Final report on project achievements	In preparation; agency submission scheduled for September 2026

Milestones

MS1–MS7 were achieved. MS7 is supported by the institutionally submitted APVV-25-0314 (OptiAIM) application, in which Michaela Horváthová is a named research-team member. The application documentation is retained privately in the project records.

4. Main Scientific Outputs

Supervised Learning for Robust Predictive Control: Safe and Tunable Approach

M. Horváthová, K. Kiš, M. Klaučo, J. Oravec. Neurocomputing 671 (2026), 132637.

[Official record](#) · [Open-access copy](#)

Experimental Validation of Deep Koopman MPC for Real-Time Pasteurization Unit Control

P. Valábek, M. Horváthová, G. Pannocchia, M. Klaučo. Control Engineering Practice 167 (2026), 106663.

[Official record](#) · [Open-access copy](#)

Variance-Adaptive Approximated Model Predictive Control

M. Horváthová, Y. Jiang, J. Holaza, S. Olaru, J. Oravec. European Control Conference 2026, pp. 523–529.

[Official record](#) · [Open-access copy](#)

Deep Koopman Economic Model Predictive Control of a Pasteurisation Unit

P. Valábek, M. Horváthová, M. Klaučo. European Control Conference 2026, pp. 19–24.

[Official record](#) · [Open-access copy](#)

Experimental Application of Stochastic Approximated MPC

Conference contribution documenting experimental use of stochastic approximated MPC.

[Official record](#) · [Open-access copy](#)

Additional dissemination output

Embedded Tube MPC with Time-Varying Cross-Section: Design and Real-Time Validation was submitted to the American Control Conference 2027. The paper documents the implementation-focused MPTplus contribution and experimental validation on a laboratory heat-exchanger platform.

5. Open-Source Software and Reproducibility

ACTREP Variance-Adaptive MPC

The dedicated repository provides the solver-free MATLAB implementation, a reproducible quadrotor example, support-controller code, tests, documentation, an example result, citation metadata and an MIT licence. The online implementation does not require MPT3, YALMIP or an external optimization solver.

Repository: https://github.com/Michaela32/Variance_Adaptive_MPC_ACTREP

Getting-started tutorial: https://github.com/Michaela32/Variance_Adaptive_MPC_ACTREP/blob/main/docs/getting_started.md

MPTplus Contribution

The project contributed a software implementation of time-varying cross-section tube MPC to MPTplus. The contribution covers implementation, automated controller synthesis, documentation, numerical verification and experimental validation; it does not claim authorship of the original underlying theoretical formulation.

MPTplus repository: <https://github.com/holaza/mptplus>

Time-varying tube MPC tutorial: <https://github.com/holaza/mptplus/wiki/Time%E2%80%90Varying-Tube-MPC-design>

6. International Cooperation and Research Impact

ACTREP strengthened collaboration with researchers and institutions including EPFL, University of Pisa, Karlsruhe Institute of Technology, Czech Technical University in Prague and other European control groups. Joint publications and conference participation expanded the project's scientific reach and supported exchange in learning-based, robust and computationally efficient MPC.

The project advanced the researcher's profile in certified learning-based control, solver-free MPC, Koopman-based predictive control and open-source tool development. Its outputs provide reusable methods, software and educational material for continued research and follow-up proposals.

7. Follow-up Grant Activity

The follow-up APVV-25-0314 application, Integrated Optimization and Machine Learning for Reliable Modelling, Monitoring, and Control (OptiAIM), was submitted on 6 February 2026 by FCHPT STU. Michaela Horváthová is a named member of the proposed research team. Because FCHPT STU is the applicant organization and the statutory representative is identified in the submitted application, Faculty-level institutional involvement formed part of the formal submission process. The application itself is not reproduced in this public report.

8. Open Science and Public Access

Project outputs are made available through publisher pages, IEEE Xplore, Zenodo, arXiv and public GitHub repositories. Official publication records are treated as authoritative citations; repository copies support open access and long-term availability. Public project documents, software, tutorials and presentations are indexed on the ACTREP GitHub page.

ACTREP project repository: <https://github.com/Michaela32/ACTREP---Fellowship-for-Excellent-Researchers>

9. Conclusion and Reporting Status

ACTREP met its core scientific and technical objectives by combining new control-method development with certification, computational simplification, experimental validation and open-source dissemination. The resulting publication portfolio and software outputs demonstrate a coherent progression from method design to practical and reproducible implementation.

This document is the public final technical summary. It is not a substitute for the formal administrative report required by the grant agency. The official report will be submitted according to the agency's reporting deadline in September 2026, after the project implementation period ending on 31 August 2026.

Public Resources

ACTREP project page: <https://github.com/Michaela32/ACTREP---Fellowship-for-Excellent-Researchers>

Variance-Adaptive MPC software: https://github.com/Michaela32/Variance_Adaptive_MPC_ACTREP

MPTplus toolbox: <https://github.com/holaza/mptplus>

UIAM project information: https://www.uiam.sk/?menu=3&show_id=5&r_p_id=180

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